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B.Tech. Degree V Semester Regular/Supplementary Examination January 2023

ME 19-205-0504 THERMAL ENGINEERING (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Understand fuels, calorimetry, and basics of combustion of fuels
 CO2: Determine the Air-Fuel ratio and analyse adiabatic flame temperature by applying thermodynamic laws to combustion.
 CO3: Study the air standard cycles used in various thermal equipments, its thermodynamic analysis, actual working of Internal Combustion Engines and its performance analysis.
 CO4: Understand various systems in Internal Combustion Engines components and alternate potential engines and their working
 CO5: Study the different phases of combustion in Internal Combustion (IC) engines and also to understand the occurrence of combustion related phenomenon in IC engines

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze,

L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A(Answer *ALL* questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) Compare solid, fluid and gaseous fuels.	3	3	L1	1	1
(b) Explain Adiabatic flame temperature.	3	3	L2	2	1
(c) Explain actual valve timing diagram of a 4 stroke petrol engine.	3	3	L2	3	1, 12
(d) Explain morse test on IC engines.	3	3	L2	3	1, 12
(e) Explain the battery ignition system of a 6 cylinder 4-Stroke petrol engine.	3	3	L2	4	1, 12
(f) Explain stratified charged engines.	3	3	L2	4	1, 12
(g) Explain Performance number (PN) and HUCR in IC engine combustion.	3	3	L2	5	1, 12
(h) Explain Cetane number and additives used in diesel fuel.	3	3	L2	5	1, 12

PART B

(4 × 12 = 48)

- II. (a) Explain Bomb calorimeter. 4 L2 1 1
 (b) The following particulars refer to an experimental determination of the calorific value of a sample of coal containing 88% carbon and 4.2% Hydrogen. 8 L3 2 2,12
 Weight of coal = 0.848 g, weight of fuse wire = 0.027 g, calorific value of fuse wire = 6700 J/g, observed temperature rise = 3.06°C, cooling correction = +0.017°C.
 Find lower and higher Calorific value of the coal.

OR

- III. (a) Differentiate between HCV and LCV of fuels. 4 L2 1 1
 (b) A single cylinder engine was supplied with a gas having the following 8 L3 2 2,12
 percentage volumetric analysis. CO = 5, CO₂ = 10, H₂ = 50, CH₄ = 25, N₂ = 10. The percentage volumetric analysis of dry flue gas was CO₂ = 8, O₂ = 6 and N₂ = 86. Determine the fuel air ratio by volume

(P.T.O.)

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		Marks	BL	CO	PO
IV.	(a) Explain actual cycle of four stroke diesel engine. Give reasons for deviation from air standard diesel cycle.	4	L2	3	1,12
	(b) The compression ratio for a single cylinder engine operating on dual cycle is 9. The maximum pressure in the cylinder is limited to 60 bar. The pressure and temperature of air at the beginning of the cycle are 1 bar and 30°C. Heat is added during constant pressure process up to 4 percent of the stroke. Assuming the cylinder diameter and stroke length as 250 mm and 300 mm respectively. Determine: (i) the air standard efficiency of the cycle (ii) The power developed if the number of working cycles are 3 per second. Take for air $c_v = 0.71$ kJ/kg K. and $c_p = 1$ kJ/kgK.	8	L3	3	2,12
OR					
V.	(a) Explain objectives and methods of scavenging.	4	L2	3	1,12
	(b) In a trial on a single cylinder oil engine working on dual cycle, the following observations were made. Compression ratio = 15, Oil consumption = 10.2 kg/h. Calorific value of the fuel = 43890 kJ/kg, Air consumption = 3.8 kg/min, Speed = 1900 rpm. Torque on the brake drum = 186 Nm, Quantity of cooling water used = 15.5 kg/min, Temp rise = 36°C, Exhaust gas temperature = 410°C, Room Temperature = 20°C, c_p for exhaust gas = 1.17 kJ/kg K, calculate: (i) Brake Power (ii) Brake Specific Fuel Consumption (iii) Brake Thermal efficiency. Draw heat balance sheet on minute basis.	8	L3	3	2,12
VI.	(a) Explain wet sump and dry sump lubrication systems with neat sketches.	6	L2	4	1,12
	(b) Explain the working of a diesel fuel pump with neat sketches.	6	L2	4	1,12
OR					
VII.	(a) Explain super charging and its effects on CI engines. Explain Indicator diagram of a supercharged CI engine.	6	L2	4	1,12
	(b) Explain with neat sketches Free piston engines and Wankel engines.	6	L2	4	1,12
VIII.	(a) Explain pre ignition in SI engines.	4	L2	5	1,12
	(b) Explain Combustion chamber designs in SI engines.	8	L2	5	1,12
OR					
IX.	(a) Explain ignition lag and factors affecting ignition lag in CI engines.	4	L2	5	1,12
	(b) Explain the three phases of CI engine combustion and factors affecting combustion in CI engines.	8	L2	5	1,12

Bloom's Taxonomy Levels

L1 = 2.5%, L2 = 70.83%, L3 = 26.77%.

